Refinement vs. Replanning: Inference Compute Allocation in Robot Policies

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Abstract

A frozen chunked robot policy can spend deployment-time inference compute in two coupled ways: more work within each policy call to refine a predicted action chunk, or more frequent policy calls to replan from fresh observations. We study this allocation without updating model weights. Our evaluation procedure reports success with both nominal compute rate and complete attempt cost, selects comparisons on discovery tasks, and tests prespecified comparisons on held-out tasks. Comparisons added after inspecting the held-out results only summarize observed patterns. We demonstrate the procedure with one frozen SmolVLA checkpoint in synchronous LIBERO simulation, using denoising count as the refinement control and fixing chunk length at $K = 50$. Arithmetic and latency characterize cost differently: at $d = 10$, 82.1% of counted arithmetic is independent of denoising, whereas a descriptive affine fit to fixed-input host time assigns 17.7% to its intercept. At $h = 2$, $d = 2$ succeeds on 139/160 attempts versus 122/160 for $d = 5$ ($+10.6$ percentage points; task-conditional 95% interval: $+4.4$ to $+16.9$), while using 5.9% less arithmetic per control step. Executing $h = 5$ rather than $h = 2$ at $d = 2$ yields 138/160 versus 139/160 successes and 60.7% lower arithmetic per attempt; equivalence is not established. The case study shows why deployment adaptation must evaluate generation effort and feedback frequency jointly; it does not define a universal setting.

1 Introduction

We can adapt a robot policy after training without changing its weights. At deployment, a frozen chunked policy has two coupled compute decisions: how much work each policy call spends generating an action chunk, and how often to call the policy again from a fresh observation. We call the first decision *refinement* and the second *replanning frequency*. Each policy call predicts K actions. The environment executes the first h, then the policy receives a new observation and runs again. Thus, h sets the policy-call frequency: one call every h executed actions. For SmolVLA, d sets the number of denoising steps within each call.

These choices couple computation and feedback. Additional refinement can alter the predicted actions. A longer execution prefix amortizes a policy call, but it delays feedback and exposes later actions from an older prediction to the environment. Holding predicted chunk length fixed, we ask one allocation question: *under an explicit compute ceiling, how much computation should each SmolVLA policy call spend on denoising refinement, and how often should the policy run again from a fresh observation?* We select the combination that maximizes closed-loop task success. Here, "best" means the highest empirical success rate among configurations that satisfy the stated compute ceiling. Counted arithmetic determines feasibility; we report nominal compute rates and realized attempt costs separately. We also report measured host latency, but it does not determine feasibility. We do not combine success and cost into one score or measure memory use.

We study one frozen SmolVLA checkpoint [1] and fix predicted chunk length at $K = 50$. Our contribution is a controlled study of the allocation between denoising work per policy call and policy-call frequency, together with its effect on task success and complete attempt cost. We do not claim that the selected allocation transfers to other checkpoints.

SmolVLA also provides an asynchronous inference mode that overlaps action generation with action execution. We do not use that mode. In our synchronous evaluation, the policy finishes generating all K actions before the environment executes the first h; the policy then receives a new observation and replans. Fixing this order lets us compare refinement and replanning without the additional effects of concurrent execution, observation staleness, or buffering. The results provide a controlled baseline for a later asynchronous study, but they do not identify the best asynchronous schedule or its performance. We do not measure asynchronous throughput or physical-robot latency.

We test 12 configurations in 480 discovery attempts, then seven settings in 1,120 held-out attempts on eight tasks excluded from selection. We chose the refinement comparisons before the held-out evaluation. We added the execution-prefix comparisons after inspecting those results, so they motivate future tests rather than support a confirmed claim.

The contributions are:

- a compute-accounting formulation that connects SmolVLA's denoising work per policy call and its policy-call frequency at fixed chunk length;
- a controlled discovery and held-out evaluation that measures task success together with nominal compute rate and complete attempt cost; and
- evidence that most counted per-call arithmetic is independent of denoising, that fewer denoising steps improve held-out success in the primary comparison, and that success under less frequent replanning remains unresolved.

2 Related Work

Post-training allocation under a compute budget. Compute-optimal scaling studies ask how a fixed training budget should be divided between model size and data [2]; related embodied work studies scaling in agent pretraining and world models [3]. Including deployment demand can change the preferred allocation [4]. These studies establish the need to state the budget and counting boundary. We instead adapt the inference configuration of one frozen policy and evaluate each allocation of deployment-time inference compute by closed-loop task success.

Computation used to generate an action. Diffusion Policy generates action sequences through iterative conditional denoising [5]. RoboMonkey spends additional inference compute on action candidates and verification [6]. This is one form of refinement: both proposal and verification cost belong to the policy call. SmolVLA provides the flow-matching policy used in our case study and also studies asynchronous inference [1]. VLA-Perf examines how model architecture, hardware, placement, networking, and asynchronous execution affect VLA inference performance [7]. These works show that arithmetic and elapsed time depend on both the model and its execution system. Our case study holds the checkpoint, hardware, and synchronous execution path fixed so that the effects of refinement and execution prefix can be compared within one system.

Choosing when to replan. ACT predicts action chunks to reduce the effective decision horizon [8]; chunked action spaces have also been studied for reinforcement learning [9]. Adaptive Action Chunking uses action entropy to choose how much of a predicted chunk to execute [10]. DVAC uses variation observed during denoising to adapt the execution prefix [11]. ElegantVLA schedules vision–language recomputation, reuse, and action generation during execution [12]. These methods change the schedule within an episode. We map fixed refinement and prefix settings first, separate selection from held-out evaluation, and report both nominal compute rate and realized attempt cost. This design evaluates axes available to an adaptive method; it does not compare a fixed deployment configuration with an online, within-episode schedule.

3 Allocating Inference Compute Between Refinement and Replanning

We represent the post-training deployment configuration of a frozen chunked policy by its model-specific refinement configuration $\mathbf{r}$, predicted chunk length K, and execution prefix h. The following quantities connect those controls to the two cost measures used in the evaluation.

At each policy query, $1 \leq h \leq K$ actions are executed and the remaining $K - h$ are discarded before replanning. For fixed or reference-calibrated call cost, let $F_{\text{gen}}(\mathbf{r}, K)$ be measured arithmetic under a stated boundary. The nominal compute rate per simulated control step is

$$b(h, K, \mathbf{r}) = \frac{F_{\text{gen}}(\mathbf{r}, K)}{h}.$$

Let $p_{\text{success}}(h, K, \mathbf{r})$ denote expected closed-loop success under the specified task distribution. For a candidate set $\mathcal{C}$ and budget ceiling B, the allocation target is

$$(h^\star, K^\star, \mathbf{r}^\star)(B) \in \arg \max_{(h,K,\mathbf{r}) \in \mathcal{C}:\, b(h,K,\mathbf{r}) \leq B} p_{\text{success}}(h, K, \mathbf{r}). \tag{1}$$

A configuration need not exhaust the ceiling. Equation 1 defines a within-checkpoint deployment decision; paired rollouts estimate its unknown success term.

In the SmolVLA instantiation, $K = 50$ and $\mathbf{r} = (d)$, where d is the denoising count. Figure 1 shows this case-study execution loop. For compactness below, $F_{\text{gen}}(d)$ denotes $F_{\text{gen}}((d), 50)$ and $b(h, d)$ denotes $b(h, 50, (d))$.

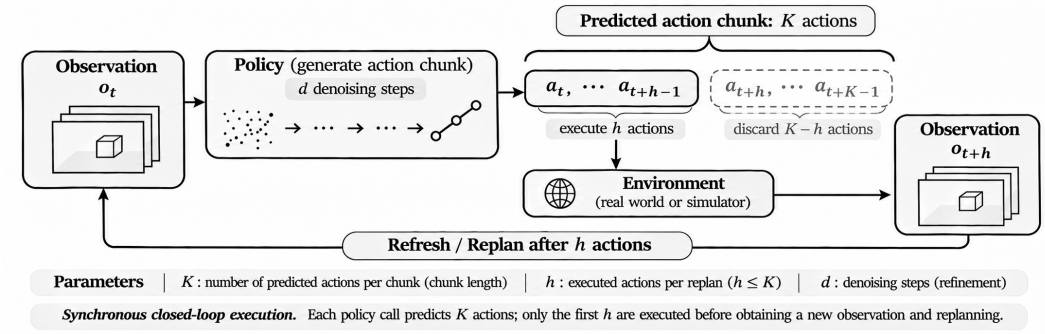


Figure 1: Closed-loop execution in the SmolVLA case study. Each policy call predicts K actions after d denoising steps, executes the first h, discards the remainder, and replans from a new observation. Here $\mathbf{r} = (d)$ and $K = 50$.

We distinguish this rate from total arithmetic per attempt. If attempt e executes T_e actions, let $N_e = \lceil T_e/h \rceil$ be its number of policy calls and let $F_{e,j}(\mathbf{r}, K)$ be the measured cost of call j. Attempt e's generated-chunk expenditure is

$$C_e = \sum_{j=1}^{N_e} F_{e,j}(\mathbf{r}, K). \tag{2}$$

The final generated chunk is charged even if only part of its execution prefix is used. For the case study, each call has fixed cost for a given d, so this reduces to $C_e = \lceil T_e/h \rceil F_{\text{gen}}(d)$. The summed form also covers variable-cost early exit, sampling, search, verification, or language generation. All-attempt means include unsuccessful episodes. Different trajectories and termination times can therefore change expenditure at the same nominal rate. FLOPs, host elapsed time, and CUDA-stream time retain separate units and measurement boundaries.

4 SmolVLA Study Design and Configuration Selection

This section applies the allocation procedure to adapt the inference configuration of one frozen checkpoint. Discovery maps the candidate settings and selects the comparisons; the held-out evaluation then tests those comparisons on tasks excluded from selection.

4.1 Discovery and held-out evaluation

We evaluate a public SmolVLA checkpoint on LIBERO [13], with

$$h \in \{1, 2, 5\}, \qquad d \in \{2, 5, 10, 20\}.$$

The predicted chunk length is fixed at $K = 50$, and the model's native deployment uses $(h, d) = (1, 10)$. We keep fixed the checkpoint, observations, preprocessing, action normalization, control period, and task-specific episode horizons.

The discovery study evaluates these 12 settings on four Spatial tasks. Each task has five initial states and two rollout seeds, giving 40 paired task–state–seed identities. Every identity receives all twelve settings in randomized order, for 480 attempts. Two discovery tasks overlap pre-study pilot work; the other two were chosen before their study outcomes were observed. This partial exposure is why discovery supports selection rather than final evaluation.

After discovery, we freeze three denoising comparisons, one at each compute ceiling. Their selected settings and comparators, plus native deployment, produce seven distinct settings. The held-out evaluation applies all seven to eight Goal and Object tasks excluded from selection. Each task has ten initial states and two rollout seeds, giving 160 attempts per setting and 1,120 in total. Prior exposure to these tasks during unrelated development is not assessed. No equivalence margin was prespecified, so similar observed success cannot establish equivalent quality. Appendix C.4 distinguishes prespecified comparisons from comparisons added after inspecting the results. The settings, identities, stopping rules, inclusion criteria, and primary analysis were fixed before held-out outcomes. After 114 attempts per primary setting, only the execution order changed; the planned identities and analyses remained fixed.

Conditions are paired by initial state and rollout seed. They also share policy noise while their queries occur at the same simulation step. Every denoising setting traverses the same integration interval; changing d changes only its discretization. Consistency checks cover initial conditions, generated and executed actions, and recorded outcomes. They do not provide an independent simulator replay; Appendix C.3 states the validation boundary.

Success gives equal weight to each task; because every task has the same number of states and seeds, this equals the overall success fraction. Descriptive 95% intervals use 20,000 bootstrap draws of states within the fixed tasks while retaining paired seeds and settings. The intervals condition on the four discovery tasks or eight held-out tasks, and secondary intervals are not adjusted for multiple comparisons. Arithmetic includes every generated chunk, including failed attempts and partly consumed final prefixes. Appendix A.3 reports timing separately.

4.2 A fixed-cost floor in the SmolVLA implementation

For the case-study checkpoint and reference input, generation cost separates into a denoising-independent term and a term that grows linearly with d. To rounding, the measured arithmetic is

$$F_{\mathrm{gen}}(d) \simeq 527.108 + 11.457\, d \quad \text{GFLOPs}$$

at all six calibrated denoising settings. Native generation at $d = 10$ costs 641.675 GFLOPs, of which 82.1% is independent of d. Halving denoising from ten to five saves 8.9% of counted generation work; reducing it to two saves 14.3%. Equivalently, relative to native deployment $(h, d) = (1, 10)$,

$$\frac{b(h, d)}{b_{\mathrm{native}}} \simeq \frac{0.821 + 0.179(d/10)}{h}.$$

Here $b_{\mathrm{native}} = b(1, 10)$. Table 1 reports the measured version of this ratio in its "Nominal rate ratio" column for all held-out settings; Appendix Table 3 reports the corresponding discovery measurements. The displayed expression is a rounded summary of those measured costs, not a separate estimate.

Profiling counts matrix products, convolutions, and attention across vision, language prefill, and the action expert, using two FLOPs per multiply-add. Embedding lookups, normalization, activations, elementwise arithmetic, and external preprocessing are excluded. We therefore refer to *dense neural arithmetic*, rather than every scalar operation. Calibration uses one real development observation and fixed tensor shapes; Appendix A.2 gives the procedure.

We prespecify compute-rate ceilings of 0.25, 0.50, and 1.00 times native. Eligibility uses measured per-d values, not the rounded affine expression. For example, $(1, 2)$ costs 0.857 times native rate and is ineligible at half-native budget, whereas $(2, 2)$ costs 0.429 times native. The denoising-independent work alone excludes $h = 1$ at half-native budget and $h \leq 3$ at quarter-native budget. A tighter budget can therefore force a longer execution prefix before any quality-based choice is made. This arithmetic floor is specific to the implementation and declared counting boundary; it is not a universal VLA law.

Calibration uses a fixed input shape. A separate audit using the language lengths observed in the tasks leaves eligibility unchanged under the same absolute ceilings (Appendix A.5). The reference-input arithmetic used for selection remains distinct from task-specific arithmetic and elapsed time. A descriptive affine fit to the six fixed-input host-time means gives $T_{\text{gen}}(d) \simeq 112.3 + 52.2\, d$ ms. Its intercept is 17.7% of fitted $d = 10$ host time, compared with 82.1% for counted arithmetic. The timing intercept is an empirical fit under one host boundary, not a component attribution. Consistently, reducing d from 10 to 5 lowers measured host time by 40.5% while counted arithmetic falls by 8.9% (Appendix A.3); FLOPs are therefore not a latency proxy. The cost model determines which settings satisfy a ceiling. The following experiments test their closed-loop outcomes.

4.3 Discovery: observed success is not monotonic in refinement

Native $(1, 10)$ succeeds in 25/40 discovery attempts. The highest observed count is 28/40 for $(1, 5)$; $(1, 2)$ and $(2, 2)$ each succeed in 26/40. At $h = 5$, success falls from 21/40 to 14/40 as d increases from 2 to 20, with intermediate counts of 18/40 and 16/40. Appendix Figure 4 shows every configuration. These point estimates do not justify imposing monotonic improvement with denoising. The small paired sample also does not certify equal quality for configurations with similar observed counts.

Selection also depends on how cost is defined. Nominal arithmetic rate differs from realized attempt expenditure, and arithmetic savings differ from elapsed time. Replacing the arithmetic ceiling with a host-time ceiling changes which prefixes are feasible. This sensitivity analysis does not alter the frozen arithmetic-based experiment. Appendix A.1 gives the full discovery results, and Appendix A.4 gives the resource-coordinate analysis.

4.4 Selection rules and prespecified comparisons

Discovery freezes $(5, 2)$, $(2, 2)$, and $(1, 2)$ at the quarter-, half-, and full-native ceilings, with $(5, 5)$, $(2, 5)$, and $(1, 5)$ as their respective comparators; $(2, 2)$ versus $(2, 5)$ is primary. Every selected setting is an observed grid point, so the held-out evaluation tests transfer to new tasks rather than extrapolation. Each pair varies denoising at a fixed prefix and compares two feasible settings; the pair does not divide an identical compute rate.

At the primary ceiling, $(2, 2)$ has the highest eligible discovery success. Among the remaining settings, $(2, 5)$ ties $(2, 10)$ and wins the lower-cost tie-break. A discovery-fitted logistic model and a simpler rule—fewest denoising steps, then shortest feasible prefix—produce the same three selections. Because the simpler rule was documented after data collection began, the agreement is descriptive and does not show an advantage from fitting. Appendix B gives the model, candidate set, and diagnostics.

5 Held-Out Success and Compute

The held-out evaluation applies seven settings to the same 160 task–state–seed identities, for 1,120 planned attempts. Three fixed-h comparisons of $d = 2$ against $d = 5$ were prespecified, with $(2, 2)$ versus $(2, 5)$ as primary. The replanning comparisons were added after the held-out results were inspected and are interpreted as observed patterns. Figure 2 shows the overall success–cost landscape; Table 1 gives the exact values. Appendix C.3 documents two removed duplicate records without changing the planned totals.

Primary result: lower refinement improves success while reducing compute. The primary comparison is designed to isolate refinement: both settings predict $K = 50$ actions, execute two actions, and then replan from a new observation; only the denoising count changes. Moving from $d = 5$ to $d = 2$ raises success from 122/160 to 139/160. Because both settings use the same task–state–seed identities, the 10.6-percentage-point increase is a within-identity difference, not a comparison between different rollout samples. The paired outcomes contain 24 cases where only

$d = 2$ succeeds, seven where only $d = 5$ succeeds, and 129 ties, giving the observed net gain of 17 successes. The task-conditional 95% interval, $[+4.4, +16.9]$, excludes zero for these eight tasks.

Table 1: Held-out results for the seven SmolVLA settings. Each setting has 160 attempts across eight tasks. The predicted chunk length is fixed at $K = 50$. Nominal rate is counted arithmetic per control step relative to native $(1, 10)$; mean arithmetic per attempt charges every generated chunk.

Allocation		Success		Nominal	Mean per attempt	
h	d	Count	(%)	rate ratio	TFLOPs	Steps
1	2	137/160	85.6	0.857	83.3	151.4
1	5	136/160	85.0	0.911	89.1	152.5
1	10	135/160	84.4	1.000	98.4	153.4
2	2	139/160	86.9	0.429	41.1	149.0
2	5	122/160	76.3	0.455	48.7	166.3
5	2	138/160	86.3	0.171	16.1	144.9
5	5	133/160	83.1	0.182	18.5	156.2

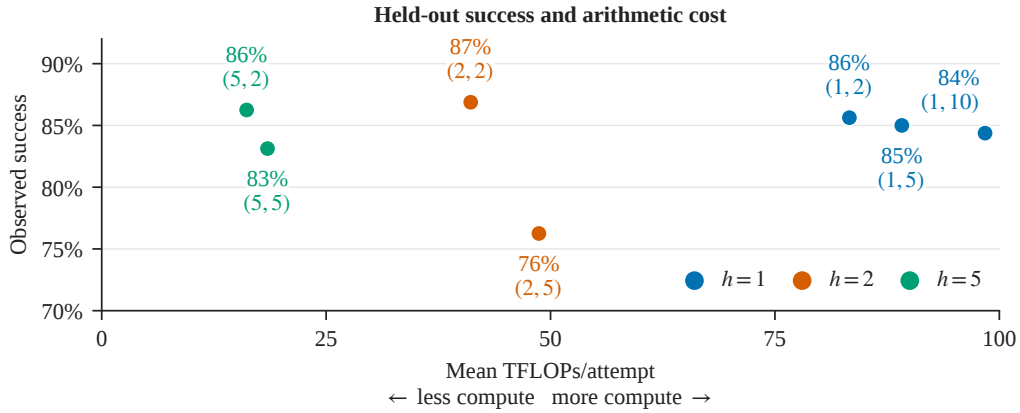


Figure 2: Held-out success against mean arithmetic per attempt for all seven settings. Every setting has 160 attempts; labels give (h, d) and observed success. Color denotes the execution prefix h; every setting uses the same circular marker. The success axis begins at 70%. Selection ceilings use nominal arithmetic per control step rather than the attempt cost on this axis (Section 4.2); exact values appear in Table 1.

The compute reduction has two scales. Lowering d reduces nominal arithmetic per control step by 5.9%. Mean arithmetic per complete attempt falls by more, 15.6%, from 48.7 to 41.1 TFLOPs, because attempt expenditure also depends on how many actions are executed and therefore how many policy calls are made. The $d = 5$ setting averages 166.3 executed steps, compared with 149.0 for $d = 2$. Thus, for this checkpoint at $h = 2$, additional denoising is both less successful and more expensive under the declared arithmetic boundary.

The result is consistent across the sampled tasks—seven favor $d = 2$ and one ties—but it is not a general denoising trend. At $h = 1$ and $h = 5$, the point estimates also favor $d = 2$, yet their intervals cross zero. The advantage may therefore depend on how refinement interacts with replanning frequency. The response model predicted the primary direction but understated its magnitude ($+4.0$ versus $+10.6$ points), so these data also do not demonstrate calibrated forecasting.

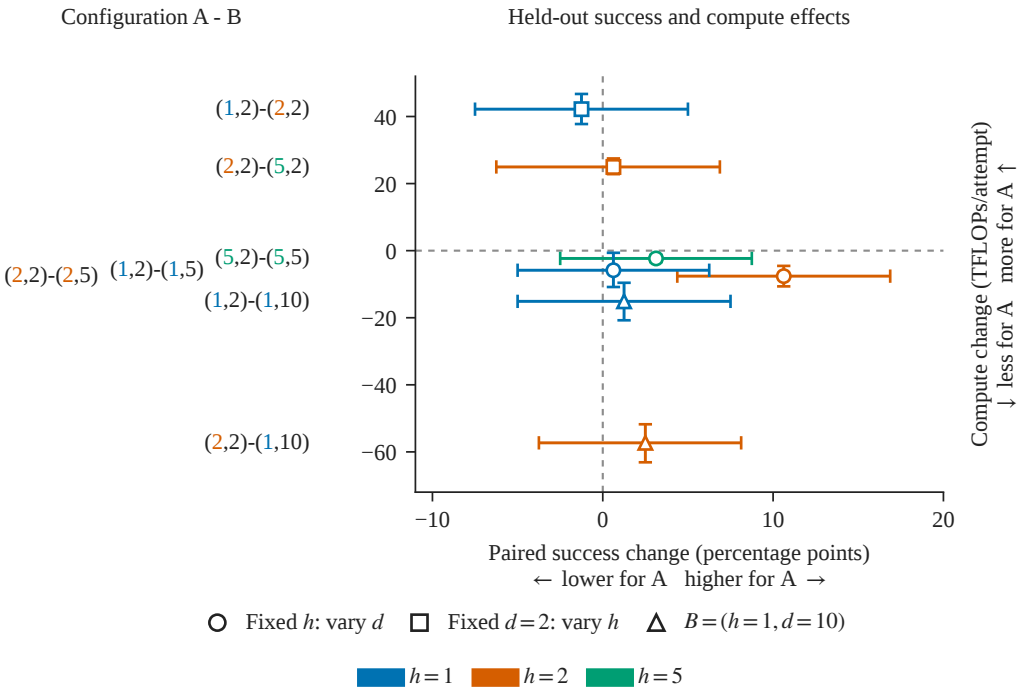


Figure 3: Joint success and compute effects for the seven held-out comparisons. In each label, the first setting is A and the second is B. The marker shows how A changes relative to B: moving right means that A succeeds more often, moving left means that it succeeds less often, moving down means that A uses less arithmetic per attempt, and moving up means that it uses more. The lower-right quadrant therefore favors A on both measures. Horizontal arms are the task-conditional 95% intervals for success change; vertical arms are the paired 95% intervals for compute change. Color denotes h for configuration A, matching Figure 2; marker shapes identify the comparison type, as shown below the plot. Exact definitions and counts appear in Table 10.

Replanning changes cost much more than observed success. At $d = 2$, executing five rather than two actions per prediction reduces mean arithmetic per attempt by 60.7%, from 41.1 to 16.1 TFLOPs. Observed success changes from 139/160 to 138/160, but the paired interval includes practically meaningful effects in both directions and task-level directions are mixed. Replanning after every action is also more expensive and shows no measured success advantage over replanning after two actions. These comparisons establish the cost differences under the declared counting boundary; they do not establish equivalent quality or a general recommendation about replanning frequency. Appendix C.1 gives their analysis status and the two contrasts involving native $(1, 10)$; Table 9 shows task-level variation.

Evidence and scope. The prespecified evidence establishes the primary result for this checkpoint at $h = 2$. The replanning comparisons were selected after inspecting the results, so their success differences require a future prespecified study. Neither varied control isolates a single mechanism: changing denoising changes numerical integration, while changing the prefix changes both observation age and executed action positions. The study uses one checkpoint, twelve tasks, and synchronous simulation; its intervals are conditional on the sampled tasks, and no equivalence margin was prespecified. Counted arithmetic excludes some operation classes and is not a latency proxy. Recorded outcomes pass source-tracing and consistency checks, but the full held-out trajectories were not replayed in an independent, version-matched simulator. Appendix C.3 defines this validation boundary. Within these limits, the held-out evidence supports lower refinement at $h = 2$ and establishes the compute—but not quality—effects of longer execution prefixes.

6 Conclusion

Adapting a frozen policy through its deployment configuration can change both success and total compute. In the primary SmolVLA comparison, reducing denoising from five to two steps improves held-out success while reducing counted arithmetic by 5.9% per control step and 15.6% per attempt. The other prespecified prefixes do not establish a uniform refinement effect. Executing five rather than two actions before replanning produces the much larger compute saving—60.7% per attempt—but the available descriptive comparison does not establish equivalent success. These results show why success, nominal rate, and complete attempt cost must be evaluated together.

Why might fewer denoising steps also improve success? One plausible explanation is that additional denoising does more than improve numerical precision: every step applies a learned update, so approximation or calibration errors can accumulate and move the first actions away from a more successful prediction. At $h = 2$, the policy executes only two of the $K = 50$ predicted actions before using a new observation, which may limit the value of further refining the full chunk. This mechanism is consistent with the primary result, but the experiment does not identify its cause. Testing it requires action-level comparisons across denoising counts and trajectory analysis around the states where outcomes diverge.

The contribution is a controlled way to study this allocation for a frozen SmolVLA checkpoint: vary denoising work and policy-call frequency, measure every call under an explicit boundary, choose comparisons before held-out evaluation, and report success with realized attempt cost. The selected setting is specific to this checkpoint and counting boundary.

The present study keeps $K = 50$ throughout, leaving predicted chunk length as a central untested allocation lever; with the tested prefixes, 90–98% of every prediction is discarded before the next query. The next study should prespecify a quality-preservation test of the cheaper $(5, 2)$ setting, vary K, and separate observation freshness from action-position effects. Repeating the same within-model design across checkpoints and before and after weight-based post-training adaptation would determine whether those adaptations also change the preferred deployment configuration. This is the evidence needed to decide when compute should refine the current prediction and when it should produce a new prediction from fresher feedback.

Data and Code Availability

Code, compact outcomes, and runnable examples can be found in the `maatinstitute/ReVa` repository. The examples reproduce the reported study and illustrate how to adapt the same evaluation to other models and checkpoints. Full simulator rollouts, model weights, and private runtime artifacts are not redistributed.

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A Discovery and Resource Measurements

Appendices A–C follow the study chronology. This section reports discovery outcomes, generation-cost calibration, fixed-input timing, sensitivity to the resource coordinate, and the language-shape audit. Appendix B documents frozen selection; Appendix C reports and verifies held-out outcomes.

A.1 Discovery outcomes

The checkpoint is `HuggingFaceVLA/smolvla_libero`, revision `6721902bc4d61e50a3bfdb11dfb4cb626f05d102`. Predicted chunk length is 50; native execution uses $h = 1, d = 10$. Discovery uses Spatial tasks 5–8, initial states 10–14, and rollout seeds 2000 and 2001. The 20 task–state blocks contain both seeds and all 12 configurations. Discovery bootstrap resampling uses 20,000 draws with analysis seed 20260913.

Table 2: Discovery successes on Spatial tasks 5–8. Each task contributes five states and two seeds per configuration. The nominal rate ratio is $F_{\text{gen}}(d)/(hF_{\text{gen}}(10))$, not realized attempt cost. The best observed count is 28/40 at $(1, 5)$; no monotonic denoising trend is assumed.

Allocation		Spatial task successes (each /10)					Nominal
h	d	Task 5	Task 6	Task 7	Task 8	Total	rate ratio
1	2	5	7	4	10	26/40	0.857
1	5	5	8	6	9	28/40	0.911
1	10	5	7	6	7	25/40	1.000
1	20	5	6	6	8	25/40	1.179
2	2	5	7	4	10	26/40	0.429
2	5	4	6	4	9	23/40	0.455
2	10	4	6	4	9	23/40	0.500
2	20	4	7	4	9	24/40	0.589
5	2	6	5	4	6	21/40	0.171
5	5	2	5	5	6	18/40	0.182
5	10	2	4	5	5	16/40	0.200
5	20	2	3	4	5	14/40	0.236

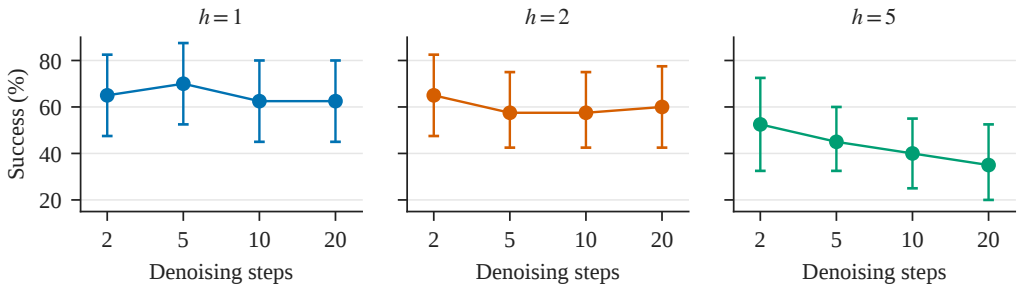


Figure 4: Discovery success by executed prefix h, with 40 attempts per configuration on four Spatial tasks. Connected estimates make the within-prefix response to denoising visible; whiskers are descriptive, task-conditional 95% paired state-bootstrap intervals. The d axes are categorical.

A.2 Generation calibration and discovery resources

Calibration uses one real observation from Spatial task 0, initial state 0, and common explicit noise. For each $d \in \{2, 3, 5, 8, 10, 20\}$, profiling records solver timesteps and dense operation counts, separately from elapsed timing. Each setting receives 20 warm-up generations, followed by three randomized timing blocks of 20 measured calls per setting. The rollout evaluator also performs 20 warm-up generations once per configuration.

Host generation timing brackets prediction and completion synchronization; CUDA events measure the current-stream prediction interval. The outer rollout clock additionally includes observation processing, action handling, simulation/rendering, and output handling. Model loading, environment creation, reset, and warm-up are measured separately. Process sampling occurs approximately every five seconds; the largest recorded gap during discovery is below 5.03 seconds. Display processes remain on the measurement GPU.

Table 3: Discovery resource measurements. Generated-chunk totals use the declared dense-arithmetic boundary and charge partially executed final prefixes. Episode means include unsuccessful attempts. $(2, 2)$ has 26/40 observed successes at 42.9% of the native nominal arithmetic rate.

Allocation		Nominal / step	Mean per attempt		Success
h	d	GFLOPs	TFLOPs	Rollout (s)	Count
1	2	550.0	90.6	39.5	26/40
1	5	584.4	91.7	61.8	28/40
1	10	641.7	109.6	111.9	25/40
1	20	756.2	129.1	199.8	25/40
2	2	275.0	45.0	21.8	26/40
2	5	292.2	53.4	38.5	23/40
2	10	320.8	57.8	61.6	23/40
2	20	378.1	67.8	107.8	24/40
5	2	110.0	20.5	12.7	21/40
5	5	116.9	23.8	20.2	18/40
5	10	128.3	27.2	32.0	16/40
5	20	151.2	33.1	55.6	14/40

Calibration requires approximately 291 seconds and discovery 31,245 seconds (8.68 hours) of host elapsed time, including setup and bookkeeping. This search expenditure is additional to subsequent deployments. The experiment does not estimate when deployment amortizes this search cost.

All measurements use an RTX 4060 Ti. Process samples taken approximately every five seconds show no unrelated sampled CUDA compute during calibration and discovery; display processes remain. Coverage is incomplete during the held-out evaluation, so its aggregate rollout times are descriptive. Host and CUDA-stream generation intervals are measured separately from the outer synchronous rollout interval, and warm-up is excluded.

A.3 Fixed-input generation timing

Table 4: Fixed-input full-generation calibration. Each setting has 60 calls in three randomized blocks. Range gives the minimum and maximum host block means, not a confidence interval. Host time falls more steeply with d than counted arithmetic. Host and CUDA columns average separate measurements of all 60 calls; the extra decimals distinguish their means, not timing accuracy.

d	GFLOPs	Host (ms)	CUDA (ms)	Host block range (ms)
2	550.0	217.891	217.862	214.5–220.1
3	561.5	269.990	269.970	265.3–272.6
5	584.4	374.321	374.300	366.8–378.1
8	618.8	529.746	529.726	519.9–536.1
10	641.7	629.624	629.604	623.2–642.2
20	756.2	1158.635	1158.608	1133.5–1172.6

The calibration stores full-generation intervals and global operator arithmetic counts, not separate vision, language-prefill, or action-expert timings and arithmetic totals. These records cannot attribute elapsed sensitivity to a particular component, memory traffic, omitted operation classes, synchronization, or software overhead. CUDA-event intervals are elapsed stream measurements, not summed active-kernel durations. No component-level decomposition is inferred.

A.4 Resource-coordinate sensitivity

Arithmetic savings and elapsed savings differ. Mean rollout time is 111.9 seconds per attempt for native deployment, 61.8 seconds for $(1, 5)$, and 21.8 seconds for $(2, 2)$. Their mean generated-chunk expenditures are 109.6, 91.7, and 45.0 TFLOPs, respectively. These are observed all-attempt means under the same task allocation; they include different trajectories and outcomes. All discovery configurations appear in Table 3.

There is also an ordering reversal: $(1, 5)$ uses more arithmetic per attempt than $(2, 20)$ (91.7 versus 67.8 TFLOPs), but less rollout time (61.8 versus 107.8 seconds). These means include different trajectories. On a fixed input, reducing d from 10 to 5 lowers mean generation host time from 629.6 to 374.3 ms (40.5%), despite only 8.9% less counted arithmetic. The calibration therefore separates arithmetic savings from elapsed savings without trajectory differences, but it does not resolve their component-level cause. FLOP and latency constraints can therefore select different settings.

As a descriptive sensitivity analysis, define calibrated host-time rate $b_T(h, d) = \overline{T}_{\mathrm{gen}}(d)/h$, the amortized fixed-input generation time per simulated control step, and normalize it by native host-time rate. For $(1, 2)$, $(2, 2)$, and $(5, 2)$, the normalized arithmetic rates are 0.857, 0.429, and 0.171, while the normalized host-time rates are 0.346, 0.173, and 0.069. Consequently, $(1, 2)$ is ineligible at a half-native arithmetic ceiling but eligible at a separately normalized half-native host-time ceiling. These resource constraints admit different replanning schedules; the choices do not establish quality optima. The frozen experiment remains arithmetic-based.

Table 5: Smallest execution prefix feasible for $d = 2$ under separately normalized arithmetic and fixed-input host-time ceilings. The resource coordinate changes feasibility; these choices do not compare outcome quality.

Ceiling relative to native	Arithmetic constraint	Host-time constraint
0.25	$(5, 2)$	$(2, 2)$
0.50	$(2, 2)$	$(1, 2)$
1.00	$(1, 2)$	$(1, 2)$

A.5 Language-shape transport audit

The profiled batch contains one sample, two $3 \times 360 \times 360$ image tensors, an eight-dimensional state tensor, and language-token and attention-mask tensors of length 20. The checkpoint exposes seven-dimensional actions and generates a 50-action chunk. Evaluation fixes batch size, image preprocessing, state/action dimensions, and predicted chunk length. The tokenizer pads language only to the longest sequence in a batch. Reconstructing the declared task strings with the pinned tokenizer gives 15–17 tokens for discovery and 5–12 for held-out evaluation, versus 20 in calibration. We profile every distinct evaluated length for $d \in \{2, 5, 10\}$, preserving all other shapes, and test eligibility against the prespecified absolute ceilings. Table 6 reports the range over evaluated lengths 5–17 against the original length-20 value.

Table 6: Dense generation arithmetic across evaluated language shapes. Budget tests retain the prespecified absolute ceilings; none of the reported held-out configurations changes eligibility.

d	Evaluated lengths GFLOPs (range)	Length 20 GFLOPs	Eligibility changes
2	539.7–547.9	550.0	0
5	573.5–582.2	584.4	0
10	629.8–639.3	641.7	0

In particular, $(2, 10)$ remains within the original half-native ceiling for all evaluated lengths; shorter inputs reduce rather than increase its counted work. The audit changes neither policy execution nor the frozen selections.

B Selection and Diagnostic Analyses

B.1 Frozen selection procedure

We compare two logistic response-model families. A compute-only model uses a quadratic in log normalized rate. An allocation model uses $\log h$, $\log(d/10)$, both squared terms, and their interaction. Each family is fitted with ridge penalty 1 or 10. Leave-one-task-out log loss selects the model, which is then refitted on the discovery set. No individual seed repeat is held out while another repeat from the same task remains in training.

The candidate set is $h \in \{1, 2, 3, 5\}$ crossed with $d \in \{2, 3, 5, 8, 10, 20\}$: the 12 discovery configurations and 12 untested configurations. Their calibrated costs determine eligibility; the fitted model selects the highest predicted success under each ceiling. Ties use compute and lexicographic order.

Table 7: Discovery-based selector choices. The learned choices and comparisons were fixed before held-out evaluation; additional selector baselines were documented after collection began. The empirical selector uses the measured grid; other selectors use the declared candidate set. The simple rule exactly matches the learned choices. The selected rate ratio is the learned recommendation's nominal arithmetic rate divided by the native rate. Prespecified comparators and predicted gaps appear in Table 8; the half-native row is primary.

	Selector choice (h, d)				Learned selector
Ceiling ratio	Learned	Simple	Empirical	Compute-only	Rate ratio
0.25	$(5, 2)$	$(5, 2)$	$(5, 2)$	$(5, 20)$	0.171
0.50 (primary)	$(2, 2)$	$(2, 2)$	$(2, 2)$	$(2, 10)$	0.429
1.00	$(1, 2)$	$(1, 2)$	$(1, 5)$	$(1, 10)$	0.857

Table 8: All prespecified denoising comparisons: selected minus comparator, holding h fixed within each row. Predicted and observed gaps and intervals are percentage points. The half-native row is primary; secondary intervals are descriptive, task-conditional, and unadjusted.

	Setting (h, d)		Success difference (pp)		
Ceiling ratio	Selected	Comparator	Predicted	Observed	95% interval
0.25	$(5, 2)$	$(5, 5)$	$+6.5$	$+3.1$	$[-2.5, +8.8]$
0.50 (primary)	$(2, 2)$	$(2, 5)$	$+4.0$	$+10.6$	$[+4.4, +16.9]$
1.00	$(1, 2)$	$(1, 5)$	$+2.2$	$+0.6$	$[-5.0, +6.3]$

B.2 Selector diagnostics

The simple rule defined in Section 4.4 reproduces every learned recommendation (Table 7); held-out outcomes cannot distinguish these selectors at the tested budgets. The compute-only choices use the original ridge-10 fit. Its quarter- and half-budget choices are absent from the held-out plan, so this evaluation cannot compare those decisions on held-out tasks.

The selected allocation model has log loss 0.7048, compared with 0.7067 for the best compute-only model. It does not beat the trivial constant-$p = 0.5$ baseline, which scores 0.6931 under the same mean natural-log binary cross-entropy; an intercept-only model fitted separately on each training fold scores 0.7170. The response-model probabilities should therefore not be read as calibrated held-out-task forecasts. The primary test evaluates a recommendation shared with simpler selectors, so it does not establish an additional benefit from model-based ranking.

C Held-Out Outcomes and Verification

C.1 Complete held-out outcomes

This subsection first reports success for every held-out task and setting, then reports the four comparisons added after collection began. Of those four, two isolate the execution prefix, one isolates

denoising, and one changes both. The held-out evaluation uses Goal and Object tasks 5–8, states 20–29, and rollout seeds 3000 and 3001. Each of its 80 task–state blocks contains both seeds. All seven configurations appear in every block.

Table 9: Held-out successes by task and configuration. Each task row has 20 attempts: ten paired initial states, each with two rollout seeds. The total aggregates 160 attempts per configuration. The primary $(2, 2)$ setting exceeds $(2, 5)$ on seven tasks and ties on one.

	$h = 1$			$h = 2$ (primary pair)		$h = 5$	
Task	$d = 2$	$d = 5$	$d = 10$	$d = 2$	$d = 5$	$d = 2$	$d = 5$
Goal 5	14	15	14	16	14	18	16
Goal 6	18	17	17	18	15	19	18
Goal 7	20	20	20	19	18	18	18
Goal 8	19	18	18	18	14	18	15
Object 5	20	19	19	20	20	20	20
Object 6	19	18	19	20	17	18	18
Object 7	9	9	10	10	7	7	8
Object 8	18	20	18	18	17	20	20
Total (/160)	137	136	135	139	122	138	133

Table 10: Post-outcome descriptive comparisons using all 160 attempts per setting. These contrasts were defined after collection began and were not prespecified or fully outcome-blind. Intervals are task-conditional, unadjusted, and do not test whether two settings preserve quality within a prespecified margin.

Setting A	Setting B	A−B (pp)	95% interval (pp)
$(2, 2)$	$(1, 10)$	$+2.5$	$[-3.8, +8.1]$
$(2, 2)$	$(5, 2)$	$+0.6$	$[-6.3, +6.9]$
$(1, 2)$	$(2, 2)$	-1.3	$[-7.5, +5.0]$
$(1, 2)$	$(1, 10)$	$+1.3$	$[-5.0, +7.5]$

The four post-outcome comparisons use 20,000 paired state-bootstrap resamples and analysis seed 20260913. Table 10 reports their success differences and uncertainty intervals, while Table 1 gives their all-attempt resource means. In particular, the $(2, 2)$ minus $(5, 2)$ expenditure difference is $+25.0$ TFLOPs per attempt, with descriptive interval $[+22.8, +27.5]$. This interval describes generated-chunk arithmetic, including the different trajectory lengths; it is not an elapsed-time comparison or evidence of equivalent quality.

C.2 Primary contrast robustness, expenditure, and runtime

The primary selected configuration $(2, 2)$ has a higher success count than $(2, 5)$ on seven tasks and ties on one (Table 9). There are 24 paired wins and seven losses. In a descriptive leave-one-task-out sensitivity analysis, the equally weighted difference over the remaining seven tasks stays positive, ranging from 9.3 to 12.1 percentage points. This post-outcome analysis does not extend inference to a new task population. The ReVa example provides the paired episode-level identities needed to reproduce the primary counts and comparison.

Table 11: Descriptive primary expenditure differences, selected minus comparator at $h = 2$, including all attempts. Intervals use the paired state bootstrap conditional on the eight tasks and were added after outcomes were observed. Timing intervals summarize sampling variation in the observed data; incomplete process-monitoring coverage remains a limitation. Differences use unrounded means and may differ from subtraction of the displayed values.

Mean per attempt	$d = 2$	$d = 5$	Difference	95% interval
Reference TFLOPs	41.1	48.7	-7.6	$[-10.6, -4.5]$
Observed rollout (s)	19.0	34.3	-15.2	$[-17.2, -13.3]$
Control steps	149.0	166.3	-17.3	$[-27.9, -6.5]$

Table 12: Runtime and evaluation conditions for the held-out evaluation.

Setting	Value
Hardware	RTX 4060 Ti, 16 GB; driver 595.84
Runtime	Python 3.12.3; PyTorch 2.7.1+cu126; CUDA 12.6; NumPy 2.2.6; MuJoCo 3.8.1
Evaluation	Batch 1; synchronous simulation; nominal 20 Hz (50 ms control period)
Episode horizons	Spatial and Object: 280 control steps; Goal: 300
Precision controls	Autocast and TF32 disabled; deterministic algorithms enabled
Attention and reuse	Eager float32 attention; within-generation KV reuse
Compilation/warm-up	Model compilation disabled; 20 warm-up generations per condition
Evaluator	Pinned evaluator revision

Runtime and environment. The available measurements do not characterize numerical precision and attention execution for every model component. Table 12 supports the stated model configuration, but not component- or kernel-level claims.

C.3 Measurement verification

The evaluation data reconcile planned identities, paired states, executed actions, and outcomes. The checks in Table 13 constrain recording and execution errors; they do not constitute an independent outcome replay in a separate simulator.

Table 13: Measurement-verification checklist.

Check	Evidence
Canonical coverage	1,120 held-out attempts and 171,805 actions; the primary subset remains 320 attempts and 50,451 actions.
Pairing and dispatch	Planned identities, paired initial states and observations, generation counts, denoising counts, and applied actions reconcile.
Excluded attempts	Two repeated completed attempts are checked and excluded; one interrupted ten-action record has no completed outcome and is excluded.
Outcome path	All 1,122 completed attempts and 172,243 steps reconcile; generated actions match simulator actions and recorded outcome fields agree.
Shared predicate	Reward and success both use LIBERO's task-goal evaluation, traced to BDDL predicates, so their agreement is a consistency check.
Native parity	One Spatial episode matches the upstream path in initial state, observation, 70-action sequence, length, and outcome.
Remaining boundary	Independent, version-matched replay and direct terminal-state inspection remain outside this audit.

C.4 Analysis status

The three refinement comparisons, paired identities, stopping rules, inclusion criteria, and primary analysis were fixed before held-out outcomes. After 114 attempts per primary setting, the remaining execution order changed, while the planned identities and analyses stayed fixed. The refinement analyses use seed 20260912. Contrasts

and diagnostics defined after collection began use seed 20260913 and are descriptive. The response model was not refitted on held-out outcomes.

Reproducibility materials. The `maatinstitute/ReVa` repository contains the reproducibility materials and runnable examples. They show how to reconstruct the reported aggregate results and how to extend the evaluation to other models, checkpoints, and simulator settings. The original study's full action records, model weights, images, and runtime artifacts are not redistributed.